AI 101 for Mechanical Engineers · Module 3

Read a table as a learning problem

Lesson 3.1

What you will learn

> Identify samples, features, targets and labels in an engineering table, then describe the shapes of the arrays used for learning.

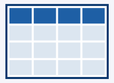
Lesson overview

Module 3 · Lesson 1

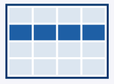
Read a table as a learning problem

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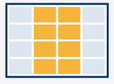
1 Know the parts of the table



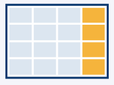
Dataset
A collection of examples and their context.



Sample
One example. Here, one steady pump test.



Features
The inputs used by the model.



Target
The output to predict: a number for regression, a category for classification.

A **label** is a supplied target value or category for an example.

2 Separate inputs from the target

Row identifier | Features: *X* | Target: *y*

Test ID	Flow (m^3/s)	Pressure rise (Pa)	Shaft angular speed (rad/s)	Measured electrical power (W)
…	…	…	…	…
…	…	…	…	…
…	…	…	…	…

The test ID keeps each row matched to its target. It is not a predictor.

Feature and target roles depend on the question.
In this task, measured electrical power is the target.

3 Check the array shapes

n = number of samples · d = number of features

X shape: (n, d)
n rows and d feature columns

y shape: $(n,)$
One scalar target per sample

120 tests · 3 features
X: (120, 3) → 120 × 3 = 360 feature values
y: (120,) → 120 target values

Shapes describe array structure, not physical units.

Key takeaway

Keep each feature row matched to the correct target and test ID.
Rows from one machine over time may be related. Keep timestamps and choose suitable evaluation splits.

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Each row is one example

- Dataset: observations and the variables measured on them [16]
- In this course, a sample means one example
- Some textbooks use “sample” for the whole set
- Pump dataset: one row is one steady operating test

> Other materials may call one row an observation or instance.

Features are inputs; the target is the output

Features

- Input variables used by a model
- Pump: flow rate, pressure rise, shaft speed

Target

- Output quantity or category to learn
- Pump: measured electrical power

A numerical target supports regression; a category supports classification.

Roles depend on the question

- A label is a supplied target annotation
- e.g. inspected bearing condition labels a vibration example
- Electrical power: target in power estimation
- Electrical power: input in fault detection

Array shapes for 120 tests and three features

Array	General shape	120 pump tests
X (features)	(n, d)	(120, 3): 120 × 3 = 360 values
y (target)	(n,)	(120,): 120 target values

n sample rows, d feature columns. Shapes describe structure, not physical units.

Keep units and rows aligned

- Flow rate m^3/s, pressure rise Pa, angular speed rad/s, power W
- Keep each variable’s meaning and unit with its column
- Same row position must describe the same test
- A test identifier is not automatically a meaningful predictor

Time-ordered rows are not independent tests

Cross-sectional records

- e.g. separate tests

Time series

- Successive readings from the same machine [18]
- Order and dependence matter
- Treating as independent can mislead evaluation

Preserve timestamps; choose splits matching the prediction or transfer question (Modules 10, 12).

CHECK YOUR UNDERSTANDING

A table contains 120 tests, three input variables and one scalar target. What are the shapes of X and y?

Answer before moving on.

CHECK YOUR UNDERSTANDING

A table contains 120 tests, three input variables and one scalar target. What are the shapes of X and y?

> X has shape (120, 3), with 360 feature values; y has shape (120,). Each target must stay aligned with its test row.

Key takeaways

- A sample is an example; features are inputs; target is output
- Feature and target roles follow the engineering question
- Array shapes separate sample rows from feature columns
- Keep features, targets and test identifiers aligned

References

[16] G. James, D. Witten, T. Hastie, and R. Tibshirani, An Introduction to Statistical Learning: With Applications in R, 2nd ed. New York, NY, USA: Springer, 2021. doi: 10.1007/978-1-0716-1418-1

[18] R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

Numbers match the course reference list. The written lesson is the authoritative text.